

Death and inhibition as morphogenetic operators:

removing tissue, stopping growth, and the gates before either enters the loop

tyssue_ops3d.Apoptosis3D · tyssue_rd_ops.Grow3D · make_sculpt.py · make_two_species.py · log/okuda/sc_*, ts_*,
tsd_* · 11 August 2026

1. The mechanism, and the arithmetic that makes it necessary

Twenty-nine rounds of this campaign, and six of the one running now, have produced spheres, buds and lobes. Not one finger. The reason is not search: Route A swept three operators over six ladders and Route B made structural edits every round. It is that **every operator in the vocabulary deforms the sheet outward**. Growth inflates, division subdivides, the purse-string necks, and the growth law is

$$\dot{v}_{\text{eq}} = \text{rate} \cdot (\rho + \text{hill}(a)), \quad \text{hill}(a) = \frac{a^n}{a_{\text{sw}}^n + a^n} \in [0, 1], \quad (1)$$

which is *purely activating*. The fastest a cell grows is set by the activator; the slowest is $\text{rate} \cdot \rho$, and $\rho > 0$ is required by premise P1 — a tissue that adds no material is not growing. **So the tissue between the spots always inflates, and a bulge can never sharpen into a finger**. Six rounds of broad lobes are what Eq. (1) predicts.

A finger requires the flanks to *stop*, and there are exactly two ways to say stop: remove the cells, or switch their growth off. Those are the two operators here.

Death is a Plexus2 family, not a convenience. The eight elementary families include *Die* — removal of biological entities — and this vesicle had no implementation of it. The biology is Monier *et al.* (2015): apoptotic force drives epithelial folding in *Drosophila*, and it is an *inward* deformation, which is the one thing nothing here could produce. Invagination is one of Okuda’s three target morphologies.

The arithmetic that decides what can be represented. An apoptotic cell leaves an epithelium by losing neighbours until it is a triangle, and a triangle collapses to a point as a T2 transition: $V - 2, E - 3, F - 1$, so $\chi = V - E + F$ is unchanged and the sheet stays closed at genus 0. Collapsing anything with $k > 3$ sides would merge k vertices into one and leave a vertex of degree k , which a trivalent sheet *cannot represent* — a rosette. So death is not a single operator: it is mark, then shed, then extrude, and the shedding belongs to **reconnect_t1_3d**. That is not a hidden coupling; it is the mechanism. A cell leaves an epithelium by losing neighbours.

2. Equations and symbols

Death, as three steps. A marked cell’s target volume decays geometrically and it is extruded when it passes a fraction of the seed-time median v_{ref} :

$$V_f^0 \leftarrow (1 - s) V_f^0, \quad \text{extrude when } V_f^0 < c v_{\text{ref}}, \quad \boxed{\tau_{\text{clear}} = \frac{\ln c}{\ln(1 - s)} \text{ ticks}} \quad (2)$$

with $s = \text{shrink_rate}$ and $c = \text{critical_frac}$. Eq. (2) is the whole of the throughput argument and it is checkable in closed form: 12 ticks at $s = 0.15$, 37 at 0.05, 94 at 0.02, 189 at 0.01, all with $c = 0.15$.

Volume is conserved into the survivors, not discarded. When cell f dies its remaining volume is shared over its live ring neighbours $\mathcal{N}(f)$:

$$c_g \leftarrow c_g + \frac{V_f^0}{|\{g \in \mathcal{N}(f) : V_g^0 \geq c v_{\text{ref}}\}| \cdot V_g^0}, \quad g \text{ live.} \quad (3)$$

The live-only restriction in the denominator is load-bearing and was found by a failure: dividing by $\max(V_g^0, 10^{-9})$ let a marked-but-undying neighbour sit at the floor and produced an activator of -1.04×10^{10} .

A death RATE, not a death set. Every state-selected rule re-evaluates each frame, so on a growing tissue marks accumulate into a wave. With N live cells,

$$|\{\text{marked}\}| \leq \phi N, \quad \phi = \text{max_mark_frac}, \quad (4)$$

and when the cap binds the *worst* candidates are taken, ranked by the mode’s own severity. The geometric modes (**band**, **cone**, **list**) are **exempt**: a set chosen once has no flux, and capping it would delete the experiment.

Inhibition supplies the zero Eq. (1) lacks, multiplicatively:

$$\dot{v}_{\text{eq}} = \text{rate} \cdot (\rho + \text{hill}(a_A)) \cdot (1 - \text{hill}_{\text{inh}}(\hat{a}_B)), \quad \hat{a}_B = a_B / \max_f a_B, \quad (5)$$

so where the second morphogen saturates the cell does not grow *at all*, baseline included. $\hat{a}_B$ is normalised to the inhibitor's own maximum for the same reason a_{sw} is a fraction: an absolute threshold against a field whose scale the chemistry sets is either always on or always off, and this project has paid for that twice.

Death — Eqs. (2)–(4)

s	shrink_rate	fraction of its target volume a marked cell loses PER TICK
c	critical_frac	extrusion threshold, a fraction OF v_{ref} (not of the cell)
v_{ref}	reference volume	the SEED-TIME median cell volume; never recomputed, or every threshold in the model moves when cells die
τ_{clear}	clearing time	ticks from sentence to extrusion. <i>Intensive</i> : a property of the pair (s, c) , not of the tissue
ϕ	max_mark_frac	fraction of LIVE cells under sentence at once. Bounds a <i>flux</i> ; the mode chooses who, this chooses how fast
n_{apop}	deaths	cumulative count. <i>Extensive</i> — it scales with the tissue, so it is never comparable between runs of different size without dividing by N

Growth and inhibition — Eqs. (1), (5)

a_A, a_B	activator of species A, B	<i>per cell</i> , in chem columns (0, 1) and (2, 3)
ρ	baseline	the growth every cell gets ungated; tip:body is $(\rho + 1) : \rho$
a_{sw}, n	gate threshold, sharpness	a FRACTION of the activator's own maximum
inhib_sw, inhib_hill	inhibition threshold, sharpness	fraction of the INHIBITOR's own maximum at which growth is half off
chan	species selector	which chem pair an operator owns. Default 0, so every pre-existing spec is unchanged

3. The Plexus2 container

No new set: death and inhibition act on sets that already exist. What is new is a *family* the vesicle did not have, and a second species inside one state buffer.

set	kind	one element is	state it carries
vertex	node	a mesh vertex	pos
cell	node	one cell of the epithelium	chem (width 4: two species), cen, area
<i>per-face mesh state, carried on the half-edge table: $V_f^0, V_{\text{birth}}, A^0, P^0$, age, ndiv, alive, apop_flag, inhib_frac</i>			

operator	family	acts on	mechanism	its gate
apoptosis_3d	Die	cell	Eqs. (2)–(4): mark, shrink, extrude as a T2	D1–D9
reconnect_t1_3d	Rewire	vertex	sheds the marked cell's neighbours — REQUIRED, or a cell shrinks forever	D4
grow_3d	Lateral	cell	Eq. (5), activation $\times$ inhibition	D11–D12
cell_react	Lateral	cell	one instance per species, chan names its pair	D10
cell_diffuse	Lateral	cell	per-column diffusivity vector	D10
seed_cell_rd	Seed	cell	seeds into its own columns	D10

Schedule, and one ordering that is not cosmetic. Death runs *between growth and the mechanics*: growth sets the targets, death overrides them for the sentenced cells, and the relaxation sees both in one tick, so a cell extruded this frame has its hole closed this frame. Measured: the same composition with death appended after the recorder instead gave grip 0.031 against 0.110, and the weakest parent in the set flipped from a collapse to a clean run.

```
seed_mesh_3d -> cell_geometry_3d -> seed_cell_rd(chan 0) -> seed_cell_rd(chan 2)
-> cell_adjacency -> cell_diffuse(0) -> cell_diffuse(2) -> cell_react(0) -> cell_react(2)
-> shape_to_chem -> grow_3d -> apoptosis_3d -> shape_energy_3d
-> interface_line_tension_3d -> reconnect_t1_3d -> divide_3d -> topo_snapshot_3d
```

There is no **substep_dt** block: this vesicle integrates at one rate, and **shape_energy_3d** carries its own **relax_iters** internally.

4. The gates

A threshold decided *before* the run, and the measured value. The tiers are the paper’s: **bookkeeping** asks whether the code does what the operator says, **closed form** whether the implementation reproduces physics it was given, and **measurement** whether the model agrees with something observed in cells. **Only the third is a statement about biology, and this note has none of it** — see §6.

gate	threshold, set before	measured
<i>tier 1 — bookkeeping: does the code do what the operator says?</i>		
D1 every mode fires on a population whose answer is known, and NOT on a uniform field	15/15 selectors	1
D2 a topology change permutes PENDING deltas, not just state	activator ≥ 0	0
D3 a dying cell’s volume goes to LIVE neighbours only	no divisor below $c v_{\text{ref}}$	1
D4 $\chi = 2$ at every frame with death running	exact	2
D5 n_{apop} equals the cells removed	exact, on a death-only composition	4
D6 the cap admits at most ϕN	8 of 100 proposed at $\phi N = 8$	8
D7 when the cap binds, the WORST are taken	the known-worst index	t
D8 the queue must DRAIN before more are sentenced	stays at ϕN	4
D9 a NAMED population is exempt from the cap	all 180	1
D10 two species never write each other’s columns	$\{0, 1\}$ and $\{2, 3\}$	e
D11 inhib_chan absent leaves an existing run BIT-IDENTICAL	exact	1
D12 growth $\rightarrow 0$ as the inhibitor saturates	$< 5\%$ of uninhibited	0
<i>tier 2 — closed form: does the implementation reproduce what it was given?</i>		
D13 clearing time equals $\ln c / \ln(1 - s)$	within 1 tick	1
D14 a collapse is a T2: $V - 2, E - 3, F - 1$	exact, else refuse	n
D15 death with no spatial pattern is EROSION, not sculpture	$\Delta \text{red_vol} < 0.05$	0
D16 a geometric chisel removes exactly its declared population	exact	5
D17 the rate cap is not the throughput lever; s is	deaths $\propto 1/\tau_{\text{clear}}$	9
<i>tier 3 — measurement: does it agree with something observed in cells?</i>		
D18 apoptotic rate matches an epithelium’s	1–3% of cells per day	1
D19 clearance time matches an extrusion in vivo	30–90 min	1
D20 the fold from a dying band matches Monier’s amplitude	—	1

Seventeen of twenty gates are tier 1 or tier 2, and three are blocked. That proportion is worth reporting rather than hiding: this note establishes that the implementation is self-consistent and that it reproduces the arithmetic it was given. **It establishes nothing about biology.**

5. The injection gates: what must hold before the loop may use this

Verification is not permission. The campaign has twice adopted an operator that was correct and unusable — **rd_interface_tension** ran for nineteen rounds at a default that could not fire, and **apoptosis_3d** destroyed every parent it touched before the rate cap existed. So the gates below are about the LOOP, not about the code, and every one is measured on the campaign’s own best parents rather than on a substrate chosen for the operator.

gate	threshold, set before	measured
I1 breaks no premise on the campaign’s own parents	0 of 6 broken	0 at $\phi = 0.005$
I2 preserves the parent’s morphology metrics	protr, grip within 10%	within 3–8%,
I3 is not inert: the acted-ledger counts it	> 0 frames	acted on all
I4 is REACHABLE: vocabulary, schedule slot, emitter	all three	menu row, SCH
I5 its default FIRES: no inert-by-construction default	mode $\neq \text{list}$	competition
I6 Route A can sweep it: a base carries it	≥ 1 ladder	6 ladders, 23 v
I7 the metric that would show its effect DISCRIMINATES	the target metric separates it from the null	see below
I8 the inhibitor’s null is bit-identical (D11)	exact	NOT RUN

I7 is the one that fails, and it is not a detail. Across all 24 death and inhibition runs, **protrusion_aspect_max_final** is 0.000 in 23 of them and 0.150 in the twenty-fourth. The loop’s own standing law L3, established independently over rounds r001–r006, says why: *aspect_peak and n_tubes do not discriminate a finger from a lobe*. So the campaign cannot currently tell whether death has produced the morphology death was added to produce. It CAN tell that death deforms — **reduced_volume** 0.807 and **invagination** 0.343 on **sc_crown** against 0.959 and 0.120 for its own control — so the honest injection is *for the deformation targets, not for the tube target*, until a finger metric exists.

6. What the runs measured, and what came back wrong

Where beats how much, by about thirty to one. `sc_crown` removed 113 cells — a polar cap, chosen once — and deformed the body more than `tsd_max` did with 3460:

run	deaths	cells	protr_f	invag_f	red_vol
<code>sc_crown</code> (polar cap)	113	16056	1.267	0.343	0.814
<code>sc_waist</code> (equator band)	332	14486	1.174	0.328	0.807
<code>sc_waist_purse</code>	336	15534	1.249	0.243	0.862
<code>tsd_max</code> (max rate)	3460	2895	1.140	0.111	0.957
control (no death)	0	12692	1.095	0.120	0.959

Scattered death erodes a ball into a smaller ball; a chisel carves. This is D15 and D16 as a positive result rather than a check.

The inhibitor works and refutes its own hypothesis. Monotone in strength — $12692 \rightarrow 6920 \rightarrow 4859 \rightarrow 4154$ cells as `inhib_sw` tightens — with `protr_final` falling $1.095 \rightarrow 1.020$. I argued in §1 that a finger needs a zero at the flanks. The zero is there, and there is no finger: inhibition makes a smaller ball, not a sharper one. The missing ingredient is therefore *not* the absence of a stop, and §1’s argument is wrong as an explanation even though it is right as arithmetic.

The two-species claim was killed by its own control. `ts_death_b_late` (death on the second map) against `ts_death_same_a` (death on the growth map): 301 vs 195 deaths, `protr_final` 1.082 vs **1.106**. The same-species control is marginally BETTER, so the second map bought nothing. That control existed to be able to say this.

Three failures worth keeping. `tsd_cap10` and `tsd_cap25` broke P4 and P12 (chemistry extinguished, then NaN), so their numbers are not evidence. `sc_old_core` — kill the oldest fastest — broke P1 and crushed the tissue to 1585 cells with negative grip: removing the body outruns growth. And **the slow-death series is confounded by my own construction**: every variant carries `inhib_sw` 0.35, which alone takes the tissue from 12692 to 4859 cells, so those five runs measure inhibition and not death speed. The question they were built to answer is still open.

7. What this licenses, and what it does not

Licensed. The Die family exists and is verified at the level of bookkeeping: selectors fire where they should and nowhere else (D1), a topology change carries its pending deltas (D2), volume is conserved into survivors (D3), the sheet stays closed (D4), the count is exact (D5), and the rate is a rate (D6–D9). Two species coexist without leaking (D10). Growth can be switched off (D12). Death patterned in SPACE deforms the body at a thirtieth of the cell cost of death patterned by rate (D15–D17), which is the result worth carrying forward.

Not licensed. No gate here is a measurement (D18–D20 blocked for want of a `units`: block, which is the same defect the `ecm` study found turned “0.82 grid cells” into $15\mu\text{m}$). The clearing time is predicted and not verified (D13). The inhibitor’s null has not been run (D11/I8), so “off by default changes nothing” is an assertion. And I7 fails: the campaign cannot yet distinguish a finger from a lobe, so it cannot be told to search for one. Injecting for the *deformation* seats — where `reduced_volume` and `invagination` do discriminate, by a factor of two against the control — is supported today; injecting for the tube seat is not.

8. References

References

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